Reproduction and Analysis of Early Prediction on Time Series Using Prefix-Based Nearest Neighbor Classification

Vivek Vishnoi

Roll No: 230119
Assignment Part B, Advanced Machine Learning

Abstract—This report reproduces and evaluates the core idea from "Early Prediction on Time Series: A Nearest Neighbor Approach" (IJCAI 2009). Using the FordA benchmark, we measure 1-nearest-neighbor accuracy across prefix lengths from 10% to 100%. The full-length baseline reaches 0.6652 test accuracy, while an early strategy at 20% prefix reaches 0.6568, indicating near-baseline performance with substantially reduced observation cost. Two ablation studies show that removing early decision slightly improves accuracy but forces maximal observation, and that distance metric choice (Euclidean vs Manhattan) materially affects accuracy at specific prefix budgets. A synthetic failure-mode experiment confirms that when early prefixes are non-discriminative, early predictions collapse before recovering once informative tail segments appear. These results validate the paper's early-vs-accuracy tradeoff and clarify the method's assumption boundaries.

Index Terms—Early Classification, Time Series, Nearest Neighbor, Prefix Prediction, Ablation Study

I. Introduction

Early classification aims to make useful predictions before observing an entire time series. This is important in time-sensitive settings where delayed prediction reduces practical value. The selected paper proposes ECTS, an extension of 1-nearest-neighbor (1NN) classification that learns when prediction is reliable from prefixes rather than waiting for full length. In this project, we reproduce the central behavioral claim on FordA using a simplified prefix-scan 1NN setup and then probe the method through ablation and failure analysis.

II. Methodology and Key Assumptions

A. Dataset and Protocol

We use the FordA dataset (3601 training, 1320 test, fixed length 500). Labels are mapped from $\{-1, +1\}$ to $\{0, 1\}$. We keep the original full length and evaluate prefix ratios from 10% to 100%; therefore, 100 timestamps are treated as a 20% prefix (not a truncated replacement dataset). The evaluation metric is test accuracy, used consistently across Tasks 2 and 3.

B. Assumptions

The method depends on three core assumptions: (1) early prefixes contain class-discriminative information, (2) nearest-neighbor distance retrieves semantically meaningful neighbors, and (3) decisions become more stable with increasing prefix length. If these assumptions are violated, early decisions become unreliable and must be deferred toward full-length observation.

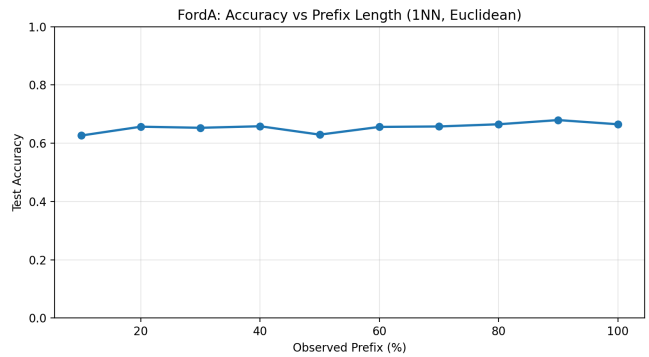

Fig. 1. FordA prefix-accuracy curve (1NN, Euclidean).

III. Empirical Reproduction

We train 1NN on each prefix ratio and evaluate test accuracy at the same observed prefix length. The full-length baseline is 0.6652. The earliest near-baseline prefix (within 1% absolute tolerance) is 20% (length 100) with accuracy 0.6568. This supports the paper's core intuition that useful decisions may be made before full sequence completion.

IV. Ablation Studies

A. Ablation 1: Remove Early Decision Mechanism

This ablation forces full-length-only prediction (no early stopping). Full-length-only achieves 0.6652, while early decision at 20% gives 0.6568. The absolute drop is 0.0083, but average observed prefix drops from 1.00 to 0.20. Thus, the early-decision component mainly contributes efficiency with a small accuracy cost.

B. Ablation 2: Change Distance Metric (Euclidean vs Manhattan)

Keeping all other settings fixed, we replace Euclidean with Manhattan distance. At full length, Manhattan

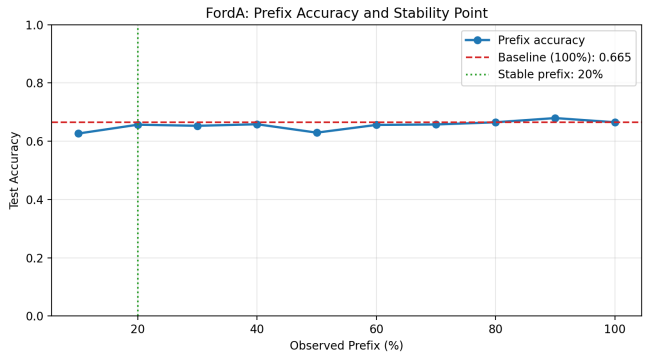

Fig. 2. Stability view with earliest near-baseline prefix at 20%.

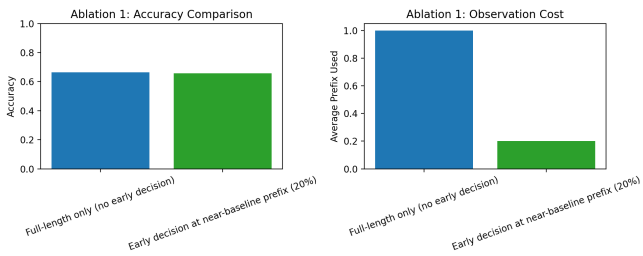

Fig. 3. Ablation 1: accuracy and observation-cost comparison.

reaches 0.6841 vs Euclidean 0.6652 (+0.0189). Across prefixes, ranking is mixed, indicating that metric selection is a meaningful secondary design choice tied to specific prefix budgets.

V. Failure Mode Analysis

To stress-test the method, we generate synthetic sequences where class prefixes are intentionally similar and only tails are discriminative. Early-prefix performance is poor (0.515 at 20%, 0.460 at 40%), then rises to 1.00 once the informative tail enters observation (60%–100%). This confirms a structural failure mode: when early evidence is non-discriminative, early nearest-neighbor predictions are unreliable even if full-length classification is strong.

VI. Reflection and Conclusion

The reproduction validates the selected paper's central tradeoff: earlier predictions can preserve most full-length performance while greatly reducing observation cost in favorable regimes. The most useful insight from this project is that early classification quality is highly assumption-dependent; once prefix informativeness breaks, performance drops rapidly. If I had more time, I would implement a confidence-based defer policy integrated into the decision rule, and evaluate it on both FordA and additional UCR datasets to test robustness beyond a single benchmark.

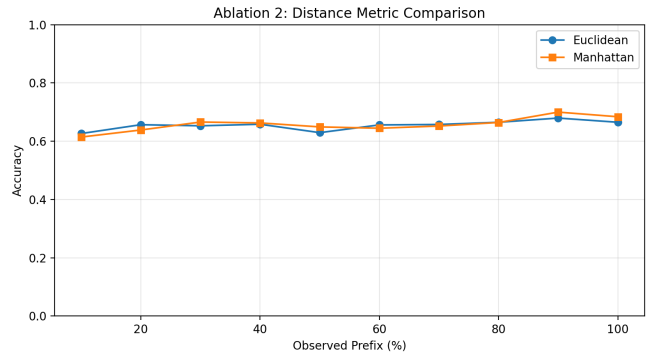

Fig. 4. Ablation 2: Euclidean vs Manhattan prefix-accuracy comparison.

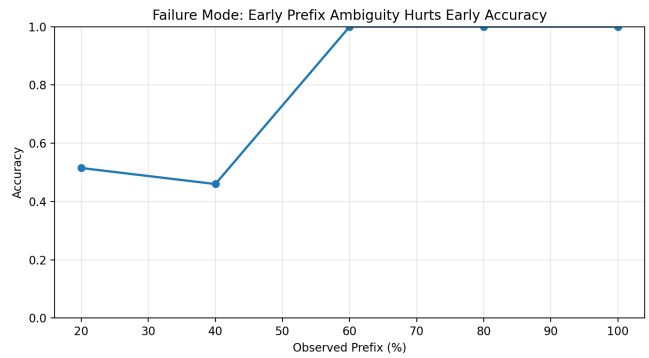

Fig. 5. Failure curve: early ambiguity hurts short-prefix accuracy.

References

[1] Y. Xiang and E. Keogh, "Early Prediction on Time Series: A Nearest Neighbor Approach," in Proc. IJCAI, 2009.

[2] H. A. Dau et al., "The UCR Time Series Archive," 2019, https://www.cs.ucr.edu/~eamonn/time_series_data_2018/.

[3] A. Bagnall, J. Lines, A. Bostrom, J. Large, and E. Keogh, "The Great Time Series Classification Bake Off," Data Mining and Knowledge Discovery, vol. 31, pp. 606–660, 2017.